



1557 - Photodissociation Region Models for JWST Observations

Cycle: 1, Proposal Category: AR

INVESTIGATORS

<i>Name</i>	<i>Institution</i>	<i>E-Mail</i>
Dr. Mark Wolfire (PI)	University of Maryland	mwolfire@gmail.com
Dr. Marc Pound (CoI)	University of Maryland	mpound@umd.edu

OBSERVATIONS

ABSTRACT

Massive O and B stars produce strong ionizing and far-ultraviolet radiation as well as powerful winds that ionize, heat, and impact their surround molecular clouds. This radiative and mechanical feedback can dramatically alter the subsequent star formation by injecting energy directly into the gas, by photodissociating molecules and ionizing atoms which alters the gas cooling, and by destroying the cloud by mechanical means or by photoevaporation. Thus, understanding the feedback process is essential to an understanding of star formation and the evolution of galaxies. The regions where this radiative energy is deposited are called Photodissociation Regions or PDRs and are key targets for JWST observations. All of the near-infrared continuum and line emission from star forming regions in nearby and low redshift galaxies to be observed by JWST arise in PDRs.

This is a Regular Archive, Theory proposal, to develop PDR models and diagnostic modeling tools to analyze the line emission from PDRs. We concentrate on the H₂ rotational emission, the CI recombination lines, the OI fluorescent lines, and the FeII fine-structure lines which reveal the gas heating rate and thermal energy content (H₂, CI, FeII), the FUV radiation field strength and penetration (H₂, OI), the H₂ formation rate (H₂), the thermal pressure (FeII), and the cloud disruption by photoevaporation (OI).

OBSERVING DESCRIPTION

